

Preflight

The preflight inspection of any airplane is critical to mitigating flight risks. A pilot transitioning into an LSA should allow adequate time to become familiar with the airplane prior to a first flight. First, the pilot and flight instructor should review the POH and cover the airplane's limitations, systems, performance, weight and balance, normal procedures, emergency procedures, and handling requirements. [Figure 17-9]

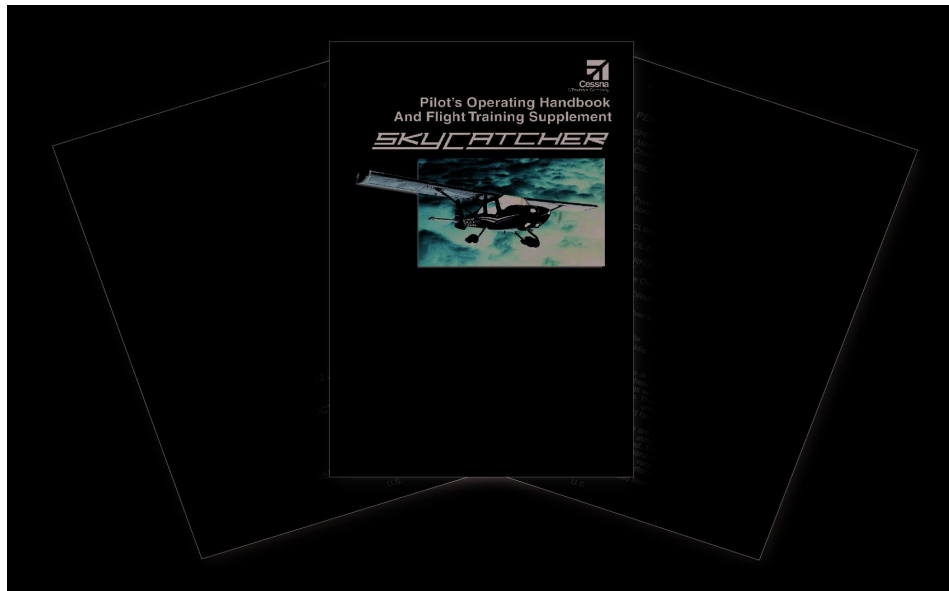


Figure 17-9. Pilot's Operating Handbook for a LSA.

Inside the Airplane

Transitioning pilots find an LSA very familiar when conducting a preflight inspection; however, some preflight differences are worth pointing out. For example, many LSAs do not have adjustable seats but rather adjustable rudder pedals. [Figure 17-10] Often, LSA seats are in a fixed position. LSA manufacturers have implemented various systems for rudder pedal position adjustment. Some manufacturers use a simple removable pin while others use a knob near the rudder pedals to adjust position. Shorter pilots may find that the adjustment range may not be sufficient, and an appropriate seat cushion may be needed to experience the proper range of rudder pedal movement. In addition, seats in some LSAs are in a semi-reclined position. The first time a pilot sits in a semi-reclining seat, it may seem somewhat unusual. A pilot should take time to get comfortable.

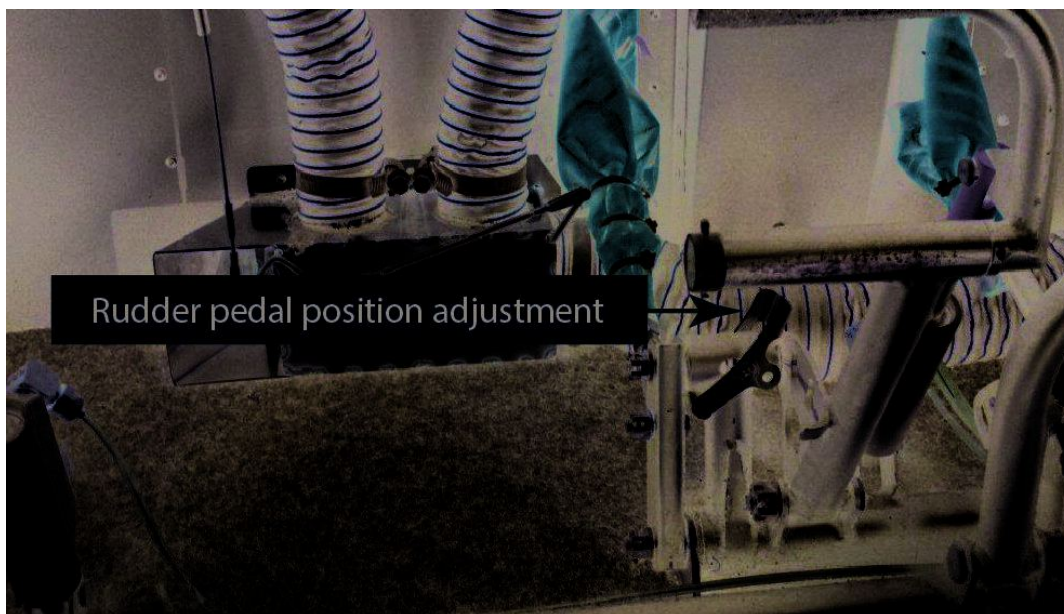


Figure 17-10. Adjustment lever for the rudder pedal position.

Transitioning pilots should become familiar with the flight and engine controls. These may vary significantly from airplane model to airplane model. Some light-sport airplanes use a conventional control stick while others use a yoke. One manufacturer has combined the two types of controls in what has been termed a “stoke.” While this control may seem unique, it provides a completely natural feel for flight control. [Figure 17-11] Regardless of the flight controls, the pilot should perform a full range of motion check of the flight controls. This means full forward to full forward left to full aft left to full aft right and then full forward right. Verify that each control surface moves freely and smoothly. On some LSAs, aileron control geometry, in an attempt to minimize adverse yaw, moves ailerons in a highly differential manner; a pilot may see very little “down” aileron when compared to the “up” aileron. Pilots should always verify the direction of control surface movement.



Figure 17-11. Stoke flight control with conventional engine controls.

Elevator trim on many LSAs is electrically actuated with no mechanical trim adjustment available. [Figure 17-12] Depending on the airplane, trim position indication may be displayed on the EFIS, an LED display, or with a mechanical indicator. On electric trim systems, as it is with any airplane, it is important to ensure that the trim position is correctly set prior to takeoff. Because trim positioning/indicating systems vary widely in light-sport airplanes, pilots should fully understand not only how to position the trim, but also how to respond to a trim-run-away condition. Part of the preflight inspection should include actuating the trim switch in both nose-up and nose-down directions, verifying that the trim disconnect (if equipped) is properly functioning, and then properly setting the takeoff trim position.



Figure 17-12. Trim control.

Depending on the engine manufacturer, the engine controls may be completely familiar to a transitioning pilot (throttle, mixture, and carburetor heat); however, some engines have no mixture control or carburetor heat. Instead, there could be a throttle, a choke control, and carburetor preheater.

Regardless, a pilot should become familiar with the specific engine installed and its operation. A transitioning pilot also needs to become comfortable with the difference between conventional engine control knobs and those found in LSAs. In standard airworthiness airplanes, control knobs are reasonably standardized; however, LSAs may use controls that are much larger or smaller in size.

If the LSA is equipped with an EFIS, the manufacturer's EFIS Pilot Guide should be available for reference. In addition, the airplane POH likely has specific EFIS preflight procedures that should be completed. These checks are to verify that all internal tests are passed, that no red "Xs" are displayed, and that appropriate annunciators are illuminated. Some systems have a "reversionary" mode where the information from one display can be sent to another display. For example, should the Primary Flight Display (PFD) fail, information can be routed to the Multi-Function Display (MFD). Not all LSA EFIS systems are equipped with a MFD or reversionary capability, so it is important for a transitioning pilot to understand the system and its limitations.

Fuel level in any airplane should be checked both visually and via the fuel level instrument or sight gauges. In LSAs, fuel level quantities can be shown using a variety of systems. Some models may have conventional float activated indicators while others may have the fuel level display on the EFIS with low-fuel alarm capability. It is not uncommon for a light-sport airplane to have advanced EFIS technology for attitude and navigation information, but have a simple sight gauge for fuel level indication. Fuel tank selection can also vary from simple on/off valves to a left/right selector. Fuel starvation remains a leading factor in aircraft accidents, which should be a reminder that when transitioning into a new airplane, time spent understanding the fuel system is time well spent.

A ballistic parachute is a popular safety feature on certain LSAs. [Figure 17-13] These devices have been shown to be well worth their cost in the unlikely event of a catastrophic failure or some other unsurvivable emergency. This system rockets a parachute into a deployed state such that the parachute slowly lowers the aircraft. The preflight inspections of these systems require a check of the mounts, safety pin and flag, and the activation handle and cable. Because most standard airworthiness type certificated airplanes do not have these systems installed, LSA training should cover the operation and limitations of the system.

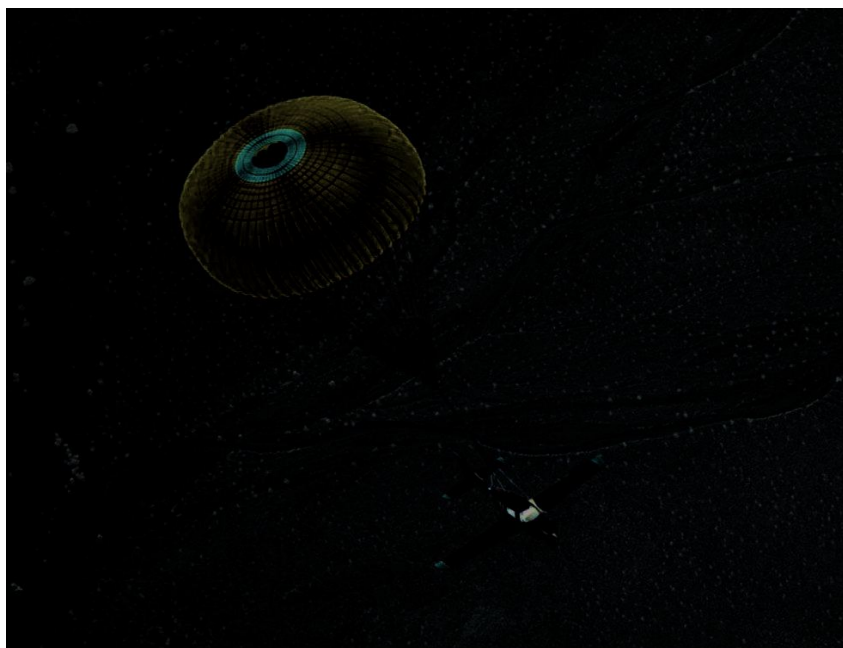


Figure 17-13. *A ballistic recovery parachute is a popular safety feature available on some LSA.*

Outside the Airplane

Transitioning pilots should feel comfortable and in a familiar setting when preflighting the outside of an LSA. Some unique areas worthy of notation are presented below.

Propellers of LSAs may range from a conventional metal propeller to composite or wood. If a transitioning pilot is principally familiar with metal propellers, time should be spent with the LSA flight instructor covering the type of propeller installed. Many LSA propellers are composite and have a ground adjustable pitch adjustment. There may be more areas to check with these types of propellers. For example, on ground adjustable propellers, ensure that the blades are tight against the hub by snugly twisting the blade at the root to verify that there is no rotation of the blade at the hub.